Crimiviz - Milestone 2

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This document outlines the design, technical requirements, and implementation roadmap for the Crimiviz interactive visualization platform, building upon the exploratory data analysis and feasibility assessments established in Milestone 1.

1 Visualization Sketches

This section presents the main structural design of our website.

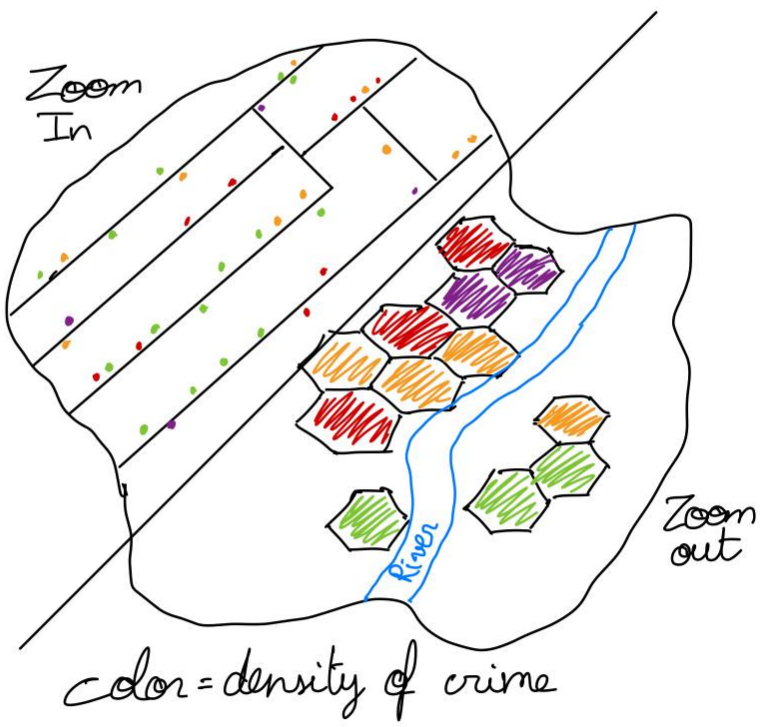

Figure 1: Sketch of the main interactive map utilizing sampling and hexbinning or heatmap to manage the density of over 8.5 million reported incidents.

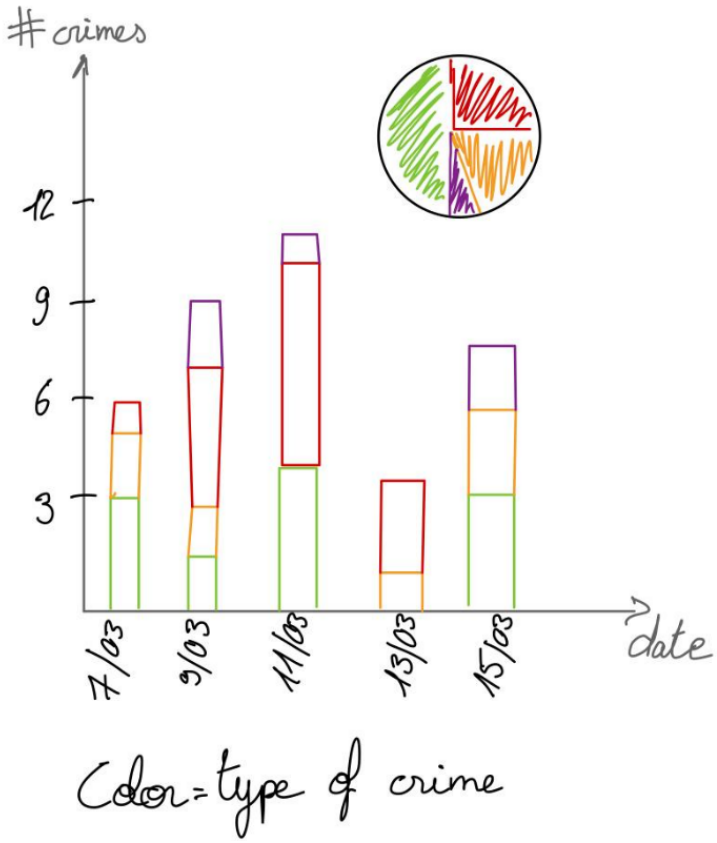

Figure 2: Sketch of a secondary visual incorporating temporal visualization, featuring interactive selectors.

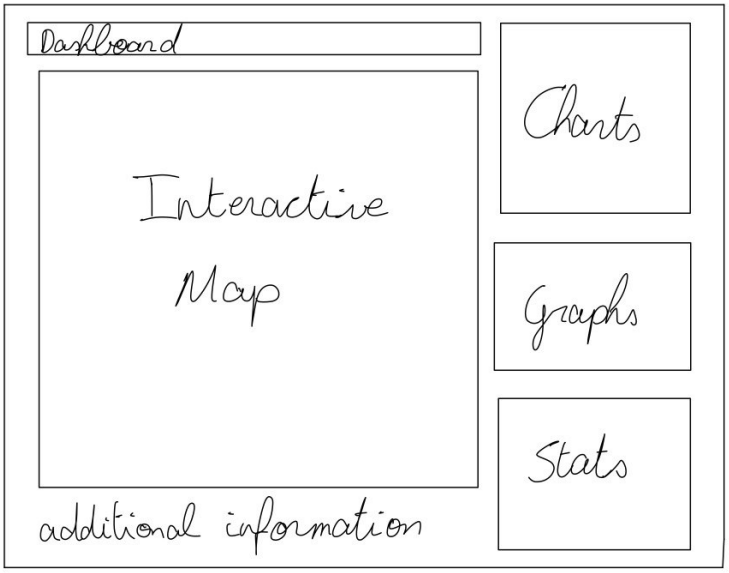

Figure 3: Sketch of the website. Just a prototype idea that needs too be refined depending on the implementations

2 Tools and Associated Lectures

List of technologies used for each visualization and the necessary course concepts required to map the spatial and temporal distributions of Chicago's criminal activities.

Visualization 1: Interactive Spatial Map (Hexbin/Sampling)

- **Tools:** D3.js for rendering, TopoJSON for mapping administrative boundaries such as Wards (50) and Community Areas (78).
- **Past lectures needed:** D3.js, Interactions, More interactive d3.js, Perception colors, Mark channels.
- **Future lectures needed:** Practical maps, Maps (for handling geospatial coordinates and boundaries with GeoJSON and TopoJSON).

Visualization 2: Temporal Trends and Arrest Rate Disparities

- **Tools:** D3.js, d3-time-format module for handling standardized datetime conversions.
- **Past lectures needed:** Data, Javascript (Part 1.1, 1.2, 2.1, 2.2), D3.js, Interactions, More interactive d3.js.
- **Future lectures needed:** Storytelling, Designing viz.

3 Project Breakdown and Minimum Viable Product (MVP)

Our required central visualization at the end, without which the project makes no sense.

3.1 Minimum Viable Product (MVP)

Our MVP will consist of:

- A functional interactive map displaying the spatial aggregation of specific crime types across Chicago's urban environments. (Visualization 1 above).
- A simple filter system allowing users to cross-filter the dataset by Primary Crime Category (e.g., Theft vs. Deceptive Practice) and the hour of the day.
- A clean but static layout that effectively communicates the city's overall threat level across a given day or year (Visualization 2 above).

3.2 Independent Pieces to Implement

To achieve this MVP and go further, we have divided the work into independent modules:

1. **Data Cleaning (Python/Pandas):** We will use a script to convert text columns into categorical variables and standardize datetimes. This process will also isolate the 1.11% of data missing precise spatial coordinates for separate evaluation.
2. **HTML/CSS User Interface:** We will set up the basic web skeleton to ensure the dashboard is visually appealing and engaging. This structured layout will house the temporal visualizations to make complex data easily understandable (See Visualization 2 above).
3. **Map Rendering (D3.js):** We will draw the foundational base map of Chicago using an optimal middle ground of spatial granularity. This main view will rely on TopoJSON boundaries to construct the 78 distinct Community Areas on an interactive map (See Visualization 1 above).
4. **Basic Interactivity (Hover Details):** We will populate the rendered map with the filtered crime data points. Adding intuitive hover functionalities will allow users to instantly display related incident information without visually overwhelming the main interface.

4 Extra Ideas

These features will improve the visualization but can be dropped if we run out of time, without compromising the project.

- **Threat Level Clock (Radial Chart):** We could implement a circular bar chart to act as a 24-hour clock. This would provide a dynamic visualization of the city's threat level, showing how crime volumes peak and dip throughout the day and maybe even link this clock to the map display.
- **Choropleth Map (Alternative View):** If mapping all 8.5 million points proves too dense, we could use an interactive choropleth map. This would group incidents by the 50 Wards or 78 Community Areas or even the 305 Police beats, coloring polygons by crime density.
- **Animated Timeline (Play Button):** To emphasize how historical crime volumes have evolved since 2001, we could add a playback control. This would animate the dataset year by year, letting users watch spatial hotspots shift across the city.
- **Real-Time Data Integration:** We could replace the static CSV file with a dynamic API connection. This would pull live updates directly from the Chicago Police Department, ensuring the visualizations constantly reflect the most current crime landscape.
- **Complex Tooltips (Sparklines):** To maximize data information density without visual clutter, we could embed mini-charts inside the map's hover tooltips. This would allow users to instantly view deeper trends and secondary metrics when exploring specific neighborhoods or locations.
- **Event-Based Overlays (Context):** We could implement contextual filters to explore specific peaks or troughs in criminal activity. This would help link crime rates to specific city events, tourism seasons, or exact hours of the day or year.

5 Website Skeleton

This section provides the link to our website skeleton.

The prototype of our website and can be accessed at the following link: `https://chicagocrime.vercel.app/`